

Multiple Choice (10 marks)

1. Which of the following conditions does not show multifactorial inheritance?
 - (a) Pyloric stenosis
 - (b) Schizophrenia
 - (c) Spina bifida (neural tube defects)
 - (d) Marfan syndrome
2. Familial hypercholesterolaemia is caused by mutations in the gene which encodes what?
 - (a) High density lipoprotein
 - (b) HMG-CoA reductase
 - (c) Low density lipoprotein
 - (d) Low density lipoprotein receptor
3. Which of the following is the most important for controlling disease caused by norovirus?
 - (a) A norovirus vaccine
 - (b) Attention to handwashing and hygiene
 - (c) Quarantine
 - (d) New antiviral drugs
4. Which of the following karyotypes is diagnostic of Down syndrome
 - (a) 46,XX,der(14;21)(q 10;q 10)pat+21
 - (b) 47,XY,+13
 - (c) 45,XX,rob,(14;21)(q 10;q 10)
 - (d) 46,XY,t(2;3)(q 21;q 12)
5. Which virus infection was not first identified by clinical observation?
 - (a) HIV
 - (b) Rubella teratogenicity
 - (c) Microcephaly and Zika
 - (d) Hepatitis C

True / False (10 marks)

Answer True or False

6. True or False: The aorta is responsible for carrying air to the lungs from the atmosphere.
7. True or False: When communicating with older adults with hearing loss, it is beneficial to face them so they can read your lips.
8. True or False: A homeotic mutation substitutes one body part for another during the development of an organism.
9. True or False: Adenoviruses used in gene therapy have a low risk of insertional mutagenesis, making them safer for long-term gene expression.
10. True or False: Recombinant alpha-iduronidase is used in the treatment of Hurler syndrome, a lysosomal storage disorder.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to a higher likelihood of depression in young survivors following a significant loss, and analyze the implications for their mental health support.
12. Discuss the significance of clonal expansion in the adaptive immune system and analyze how it contributes to the effectiveness of immune responses against pathogens.

End of Paper